DATABASE MANAGEMENT LAB Report

Submitted in partial fulfilment of the requirement of the degree of

BACHELORS OF TECHNOLOGY

to

K.R Mangalam University

Submitted by:

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Submitted to

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Assignment 2

Objective:

• To understand and implement SQL transactions using COMMIT, SAVEPOINT, and ROLLBACK, retrieve records using SELECT queries, and perform INSERT, UPDATE, and ALTER operations on tables.

Instructions:

1. Open your preferred Database Management System (DBMS) tool.
2. Create the required tables: CLASS, EMPLOYEES, and Client_master.
3. Insert the given/suitable records into each table.
4. Execute the SQL scripts and queries step by step.
5. Observe the output carefully after each operation, especially after ROLLBACK and UPDATE commands.

Table Structures:

CLASS

D (NUMBER) and Name (VARCHAR).

The initial records in the table are (1, 'Bravo'), (2, 'Alex'), and (4, 'Cheng').

EMPLOYEES

Employee_Id (NUMBER), First_Name (VARCHAR), Last_Name (VARCHAR), Email (VARCHAR), Phone_Number (VARCHAR), Hire_Date (DATE), Job_Id (VARCHAR), Salary (NUMBER), Commission_Pct (NUMBER), Manager_Id (NUMBER), and Department_Id (NUMBER).

DDL Questions (Data Definition Language):

1. How do you create a table named CLASS with appropriate data types?

2. How do you create a table named EMPLOYEES with the specified attributes?
3. How do you create a table named Client_master with the specified attributes?
4. What SQL statement would you use to rename the table Client_master to Client12?
5. How do you display the bal_due column heading as “BALANCE”?

DML Questions (Data Manipulation Language):

Part A: Transactions and Savepoints (CLASS Table)

Give the output of the following SQL script:

- INSERT INTO class VALUES (5,’Rahul’);
- COMMIT;
- UPDATE class SET name = ‘Abhijeet’ WHERE id=’5’;
- SAVEPOINT A;
- INSERT INTO class VALUES (6,’Chris’);
- SAVEPOINT B;
- INSERT INTO class VALUES (7,’Bravo’);
- SAVEPOINT C;
- SELECT * FROM class;
- ROLLBACK TO B;
- SELECT * FROM class;
- ROLLBACK TO A;

Part B: Retrieving Records (EMPLOYEES Table)

1. Find out the employee id, names, and salaries of all the employees.
2. List out the employees who work under manager 100.
3. Find the names of the employees who have a salary greater than or equal to 4800.

4. List out the employees whose last name is ‘AUSTIN’.
5. Find the names of the employees who work in departments 60, 70, and 80.
6. Display the unique Manager_Id.

Part C: Updating Records (Client_master Table)

1. Insert five records into the Client_master table.
2. Find the names of clients whose bal_due > 5000.
3. Change the bal_due of ClientNO "C123" to Rs. 5100.

DDL Questions (Data Definition Language):

1. How do you create a table named CLASS with appropriate data types?

```
CREATE DATABASE Radhika_labAssignment_2;
USE Radhika_labAssignment_2;
CREATE TABLE CLASS(
ID INT,
NAME VARCHAR(100)
);
INSERT INTO CLASS VALUES (1, 'Bravo');
INSERT INTO CLASS VALUES (2, 'Alex');
INSERT INTO CLASS VALUES (4, 'Cheng');
COMMIT;
SELECT * FROM CLASS;
CREATE TABLE EMPLOYEES (
    Employee_Id INT PRIMARY KEY,
    First_Name VARCHAR(30),
    Last_Name VARCHAR(30),
    Email VARCHAR(50),
```

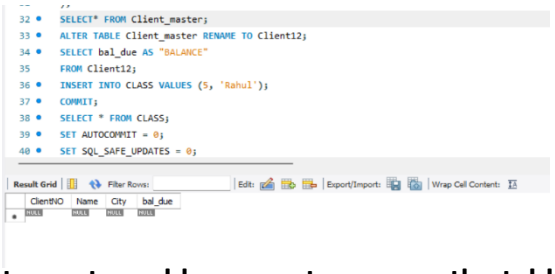
ID	NAME
1	Bravo

2.) How do you create a table named EMPLOYEES with the specified attributes?

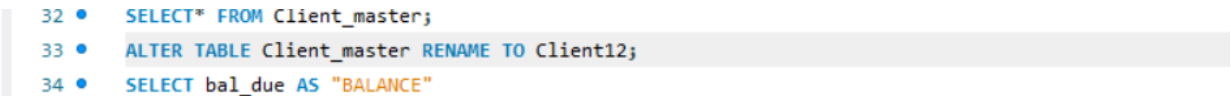
```
    Job_Id VARCHAR(20),
    Salary INT ,
    Commission_Pct INT ,
    Manager_Id INT ,
    Department_Id INT
);
SELECT* FROM EMPLOYEES;
CREATE TABLE Client_master (
    ClientNO VARCHAR(10) PRIMARY KEY,
    Name VARCHAR(30),
    City VARCHAR(30),
    bal_due INT
);
SELECT* FROM Client_master;
ALTER TABLE Client_master RENAME TO Client12;
SELECT bal_due AS "BALANCE"
```

Employee_Id	First_Name	Last_Name	Email	Phone_Number	Hire_Date	Job_Id	Salary	Commission_Pct	Manager_Id	Department_Id
NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL

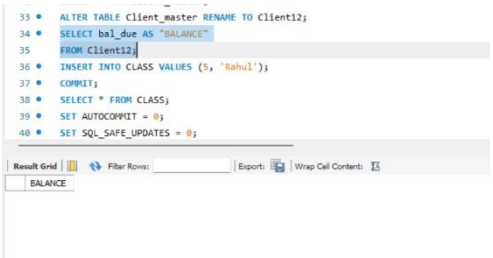
3.) How do you create a table named Client_master with the specified attributes?



4.) What SQL statement would you use to rename the table Client_master to Client12?



5.) display the bal_due column heading as "BALANCE"?



DML Questions (Data Manipulation Language):

Part A: Transactions and Savepoints (CLASS Table)

Give the output of the following SQL script:

INSERT INTO class VALUES (5,'Rahul');

COMMIT;

UPDATE class SET name = 'Abhijeet' WHERE id='5';

SAVEPOINT A;

INSERT INTO class VALUES (6,'Chris');

SAVEPOINT B;

INSERT INTO class VALUES (7,'Bravo');

SAVEPOINT C;

SELECT * FROM class;

ROLLBACK TO B;

SELECT * FROM class;

ROLLBACK TO A;

Output:

```
INSERT INTO CLASS VALUES (5, 'Rahul');
COMMIT;
SELECT * FROM CLASS;
SET AUTOCOMMIT = 0;
SET SQL_SAFE_UPDATES = 0;
UPDATE class SET name = 'Abhijeet' WHERE id='5';
SAVEPOINT A;
INSERT INTO CLASS VALUES (6, 'Chris');
SAVEPOINT B;
INSERT INTO CLASS VALUES (7, 'Bravo');
SAVEPOINT C;
SELECT * FROM CLASS;
ROLLBACK TO B;
SELECT * FROM CLASS;
ROLLBACK TO A;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

ID	NAME
1	Bravo
5	Abhijeet

```
SET AUTOCOMMIT = 0;
SET SQL_SAFE_UPDATES = 0;
UPDATE class SET name = 'Abhijeet' WHERE id='5';
SAVEPOINT A;
INSERT INTO CLASS VALUES (6, 'Chris');
SAVEPOINT B;
INSERT INTO CLASS VALUES (7, 'Bravo');
SAVEPOINT C;
SELECT * FROM CLASS;
ROLLBACK TO B;
SELECT * FROM CLASS;
ROLLBACK TO A;
SELECT* FROM CLASS;
INSERT INTO EMPLOYEES VALUES
(101, 'Khushi', 'Jindal', 'khushi@gmail.com', '9876543210', '2023-01-10', 'SOFTWARE_DEV', 12000, NULL, 100,
```

sult Grid | Filter Rows: | Export: | Wrap Cell Content:

ID	NAME
1	Bravo
5	Abhijeet

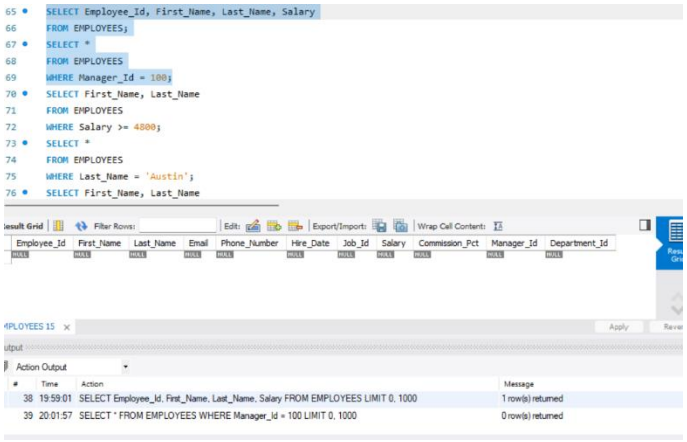
Part B: Retrieving Records (EMPLOYEES Table)

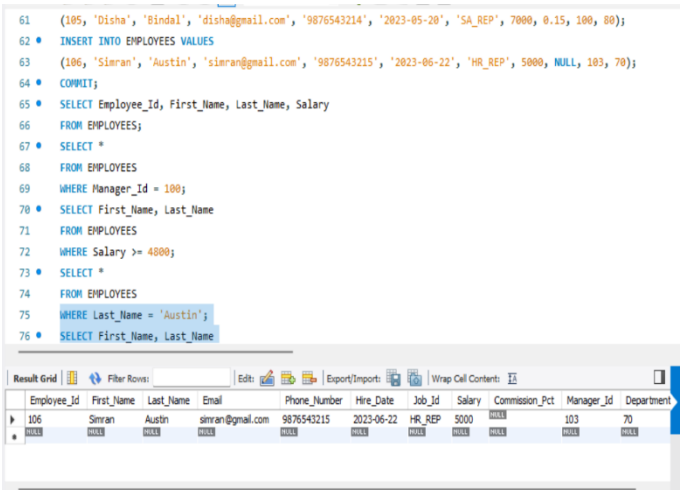
1. **Find out the employee id, names, and salaries of all the employees.**
2. **List out the employees who work under manager 100.**
3. **Find the names of the employees who have a salary greater than or equal to 4800.**
4. **List out the employees whose last name is 'AUSTIN'.**
5. **Find the names of the employees who work in departments 60, 70, and 80.**
6. **Display the unique Manager_Id.**

```
INSERT INTO EMPLOYEES VALUES
(101, 'Radhika', 'gupta', 'Radhika@gmail.com', '9876543210', '2023-01-10', 'SOFTWARE_DEV', 12000, NULL, 100, 60);
INSERT INTO EMPLOYEES VALUES
(102, 'Prachi', 'Goyal', 'prachi@gmail.com', '9876543211', '2023-02-12', 'HR_REP', 4800, NULL, 101, 70);
INSERT INTO EMPLOYEES VALUES
(103, 'subhi', 'Gupta', 'subhi@gmail.com', '9876543212', '2023-03-15', 'SA_REP', 5200, 0.10, 100, 80);
INSERT INTO EMPLOYEES VALUES
(104, 'Priyanka', 'Austin', 'priyanka@gmail.com', '9876543213', '2023-04-18', 'SOFTWARE_DEV', 4500, NULL, 102, 60);
INSERT INTO EMPLOYEES VALUES
(105, 'Disha', 'Bindal', 'disha@gmail.com', '9876543214', '2023-05-20', 'SA_REP', 7000, 0.15, 100, 80);
INSERT INTO EMPLOYEES VALUES
(106, 'Simran', 'Austin', 'simran@gmail.com', '9876543215', '2023-06-22', 'HR_REP', 5000, NULL, 103, 70);
COMMIT;
SELECT Employee_Id, First_Name, Last_Name, Salary
FROM EMPLOYEES;
SELECT *
```

Result Grid | Filter Rows: | Edit: | Export/Import: | Wrap Cell Content:

Employee_Id	First_Name	Last_Name	Salary
106	Simran	Austin	5000
NULL	NULL	NULL	NULL





Part C: Updating Records (Client_master Table)

1.) **Insert five records into the Client_master table.**

```
INSERT INTO Client12 VALUES
('C123','Sakshi Sharma','Delhi',4500);
INSERT INTO Client12 VALUES
('C124','Anita Verma','Mumbai',6000);
INSERT INTO Client12 VALUES
('C125','Kaira Singh','Jaipur',7500);
INSERT INTO Client12 VALUES
('C126','Isha Mehta','Noida',3000);
INSERT INTO Client12 VALUES
('C127','Ashi Joshi','Chandigarh',8200);
COMMIT;
SELECT* FROM Client12;
```

OUTPUT :

Result Grid | Filter Rows:

ClientNO	Name	City	bal_due
C123	Sakshi Sharma	Delhi	4500
C124	Anita Verma	Mumbai	6000
C125	Kaira Singh	Jaipur	7500
C126	Isha Mehta	Noida	3000
C127	Ashi Joshi	Chandigarh	8200
NULL	NULL	NULL	NULL

2.) Find the names of clients whose bal_due > 5000.

```
SELECT Name
FROM Client12
WHERE bal_due > 5000;
```

OUTPUT :

Result Grid

Name
Anita Verma
Kaira Singh
Ashi Joshi

3.) Change the bal_due of ClientNO "C123" to Rs. 5100.

```
UPDATE Client12
SET bal_due = 5100
WHERE ClientNO = 'C123';
COMMIT;
SELECT* FROM Client12;
```

OUTPUT :

Result Grid | Filter Rows:

ClientNO	Name	City	bal_due
C123	Sakshi Sharma	Delhi	5100
C124	Anita Verma	Mumbai	6000
C125	Kaira Singh	Jaipur	7500
C126	Isha Mehta	Noida	3000
C127	Ashi Joshi	Chandigarh	8200
NULL	NULL	NULL	NULL